



IF3170 Inteligensi Artifisial

Institut Teknologi Bandung

Adversarial Search

Semester I TA 2026/2027





Outline

- Review:
 - What is AI
 - Intelligent Agent: PEAS, Property of Task Environment
 - Local Search
- Adversarial Search



Adversarial
Search

Non-
Adversarial
Search

Approximate
solution:
strategy

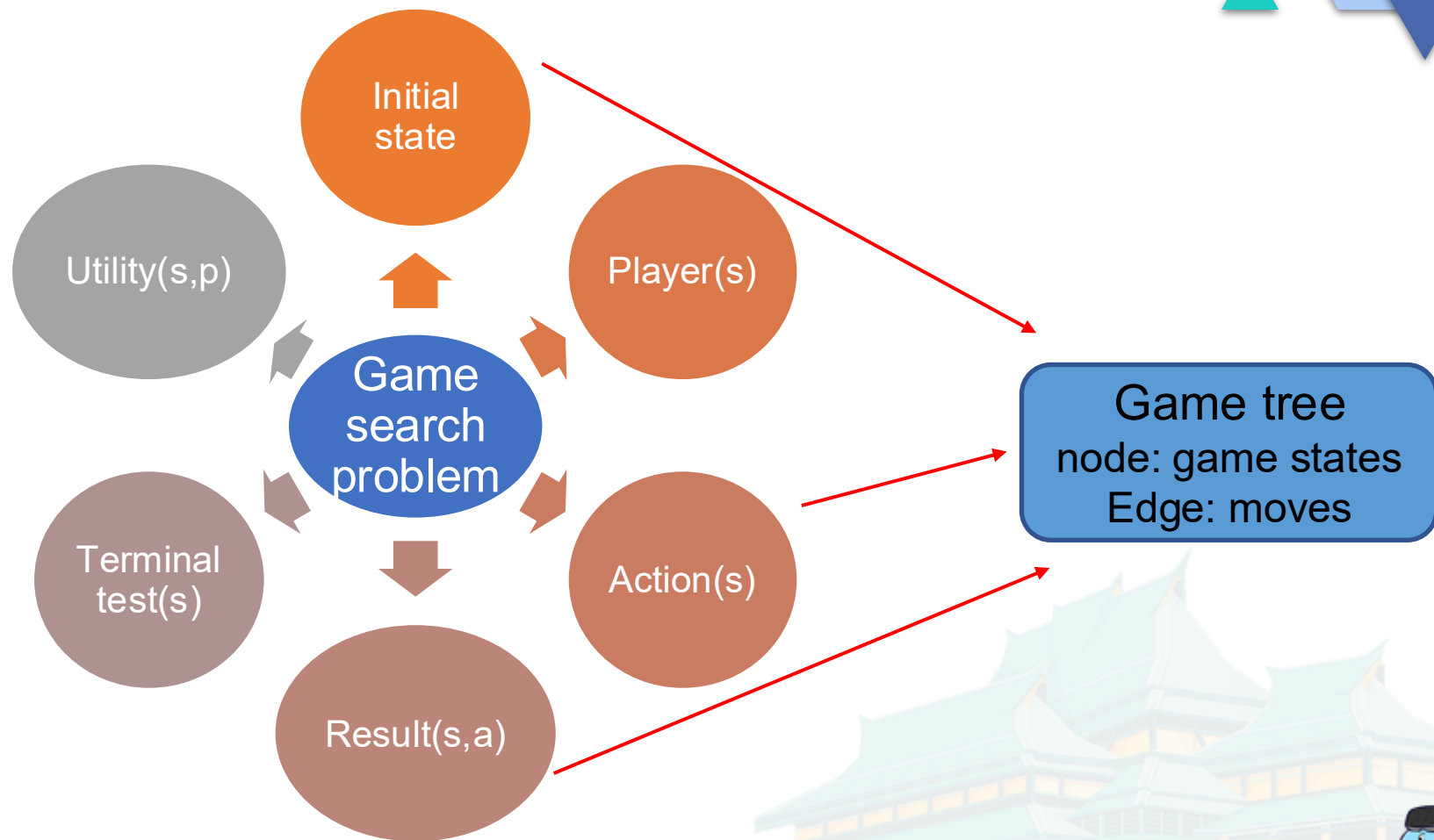
Multiagent:
competitive
environment
(games)

Optimize
solution: path
to goal or
solution state

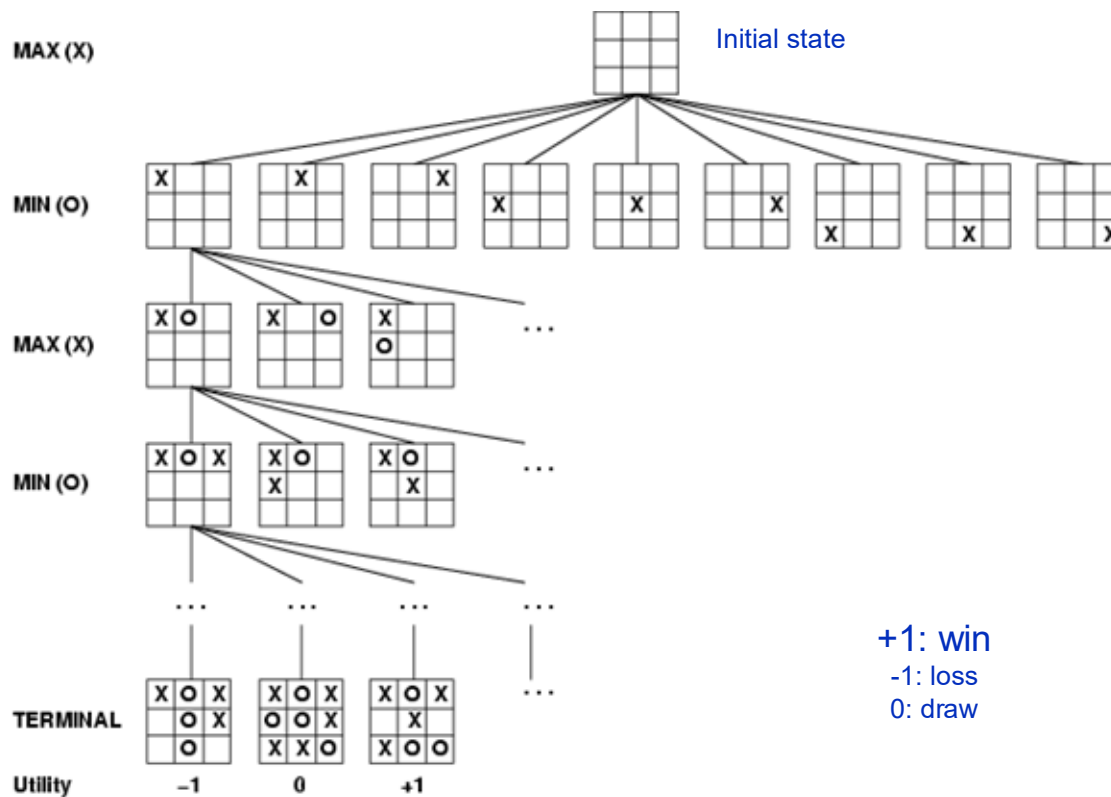
Deterministic
and fully
observable

Turn-taking





Game tree (2-players, tic-tac-toe)



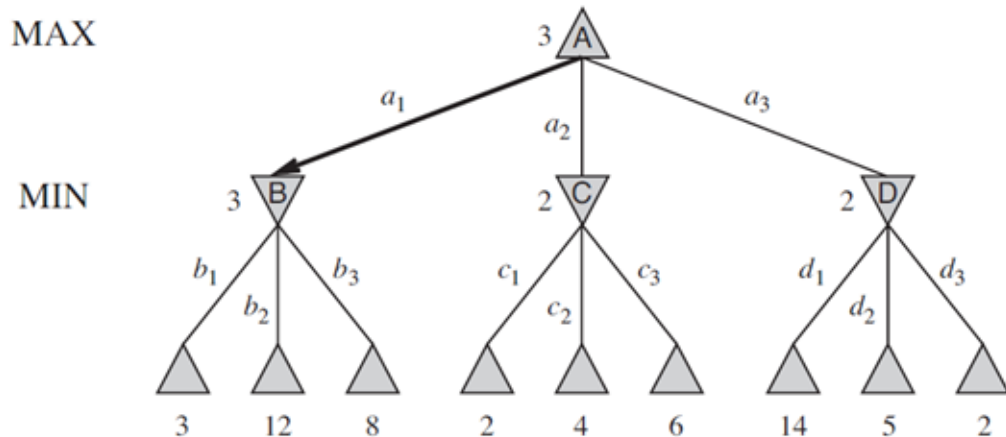
function MINIMAX-DECISION(*state*) **returns** *an action*
return $\arg \max_{a \in \text{ACTIONS}(s)} \text{MIN-VALUE}(\text{RESULT}(\text{state}, a))$

function MIN-VALUE(*state*) **returns** *a utility value*
if TERMINAL-TEST(*state*) **then return** UTILITY(*state*)
 $v \leftarrow \infty$
for each *a* **in** ACTIONS(*state*) **do**
 $v \leftarrow \text{MIN}(v, \text{MAX-VALUE}(\text{RESULT}(s, a)))$
return *v*

function MAX-VALUE(*state*) **returns** *a utility value*
if TERMINAL-TEST(*state*) **then return** UTILITY(*state*)
 $v \leftarrow -\infty$
for each *a* **in** ACTIONS(*state*) **do**
 $v \leftarrow \text{MAX}(v, \text{MIN-VALUE}(\text{RESULT}(s, a)))$
return *v*



Minimax: 2-ply Game Tree



$\text{Minimax}(B) = \min(\text{Minimax}(\text{result}(B, b_1)), \text{Minimax}(\text{result}(B, b_2)), \text{Minimax}(\text{result}(B, b_3)))$
 $= \min(3, 12, 8) = 3$

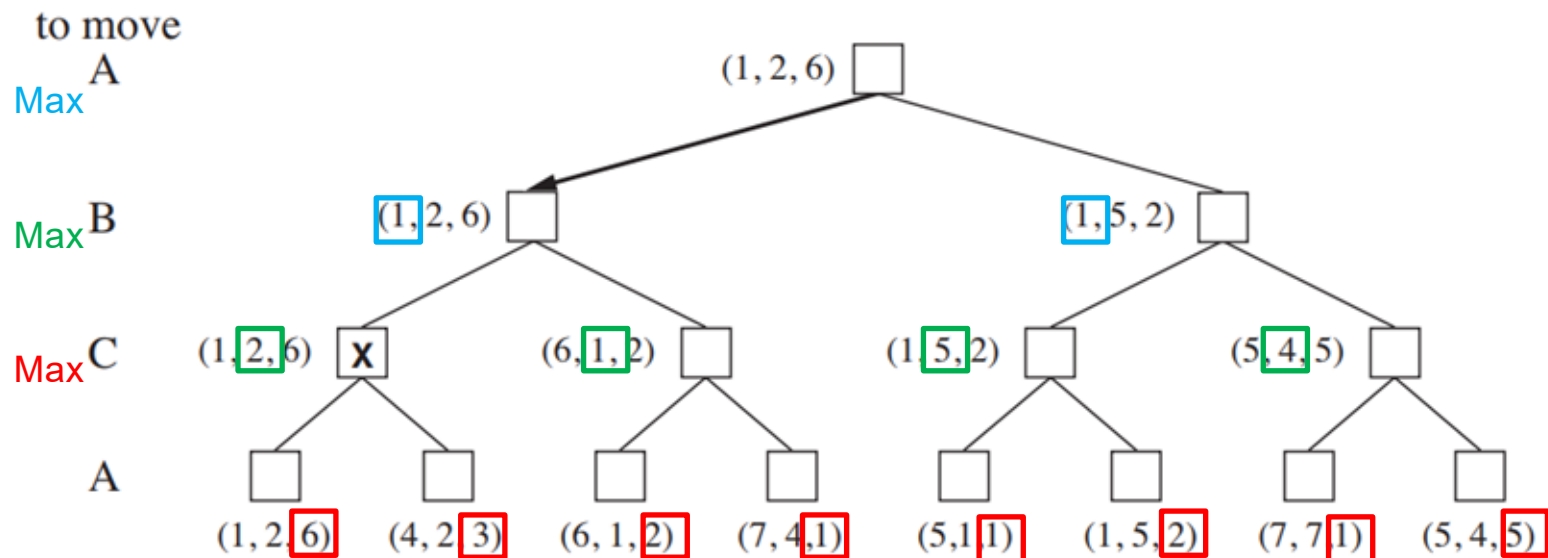
$\text{Minimax}(A) = \max(\text{Minimax}(B), \text{Minimax}(C), \text{Minimax}(D))$
 $= \max(3, 2, 2) = 3$

$\text{MINIMAX}(s) =$

$$\begin{cases} \text{UTILITY}(s) & \text{if } \text{TERMINAL-TEST}(s) \\ \max_{a \in \text{Actions}(s)} \text{MINIMAX}(\text{RESULT}(s, a)) & \text{if } \text{PLAYER}(s) = \text{MAX} \\ \min_{a \in \text{Actions}(s)} \text{MINIMAX}(\text{RESULT}(s, a)) & \text{if } \text{PLAYER}(s) = \text{MIN} \end{cases}$$



Minimax 3-ply in Multiplayer Games



Minimax Properties

Complete?

Yes (if tree is finite)

Optimal?

Yes (against an optimal opponent)

Time complexity?

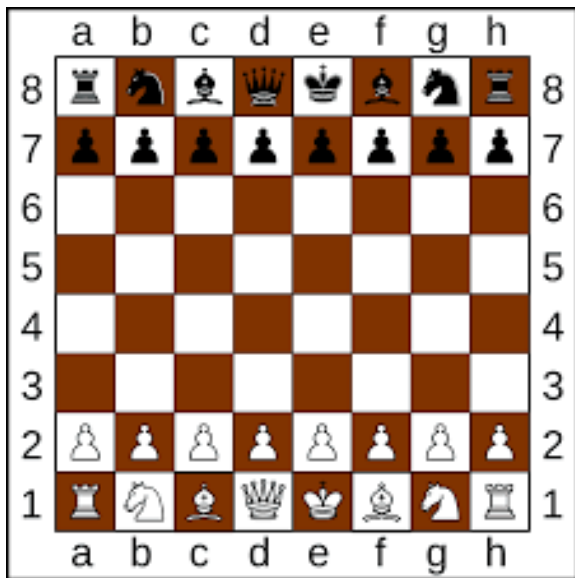
$O(b^m)$

Space complexity?

$O(bm)$ (depth-first exploration)



Minimax for Chess

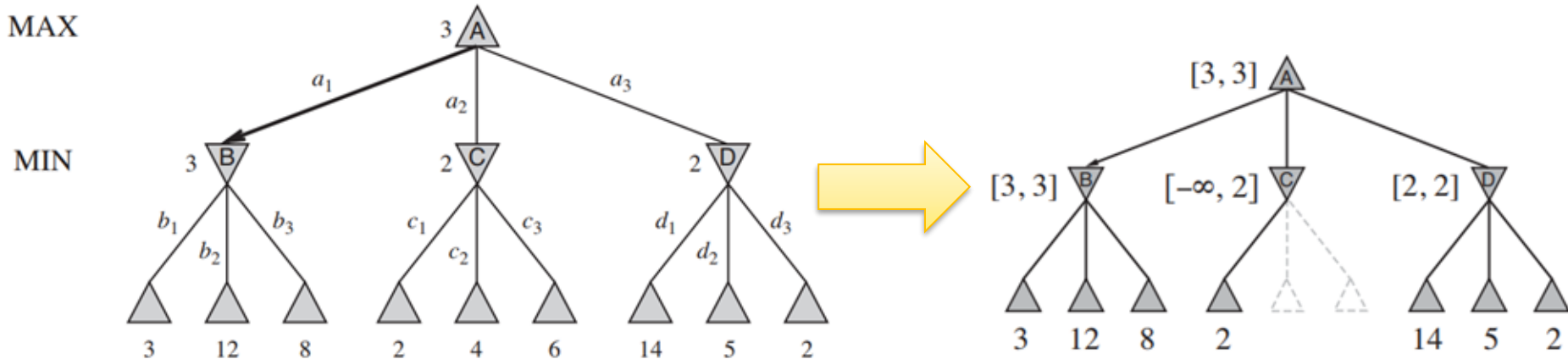


https://commons.wikimedia.org/wiki/File:AAA_SVG_Chessboard_and_chess_pieces_02.svg

- Branching factor: 35 (avg)
- Games often 50 moves for each player -> $m=100$
- Game states is exponential in the depth of game tree.
- Exact solution is completely infeasible



$\alpha\beta$ Search: Minimax with Pruning

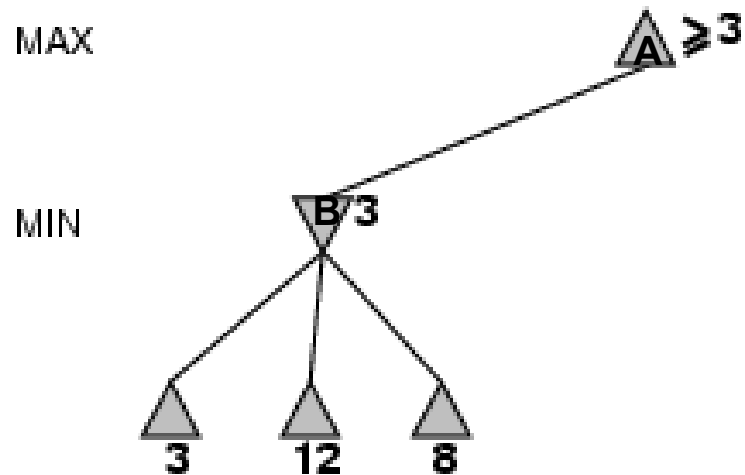


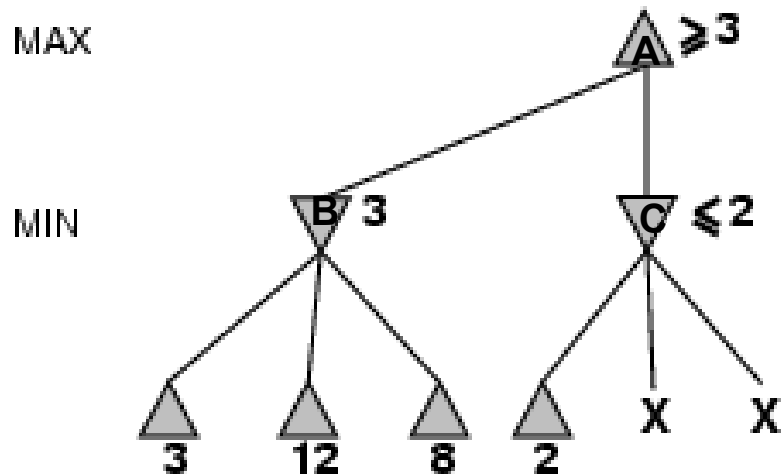
Pruning **does not** affect final result. It returns the same move as minimax would, but prunes away branches that cannot possibly influence the final decision.

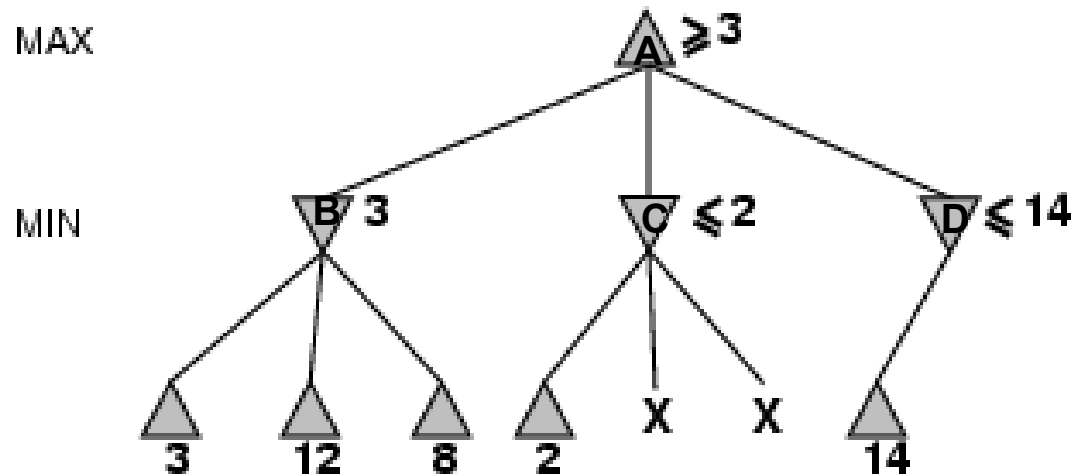


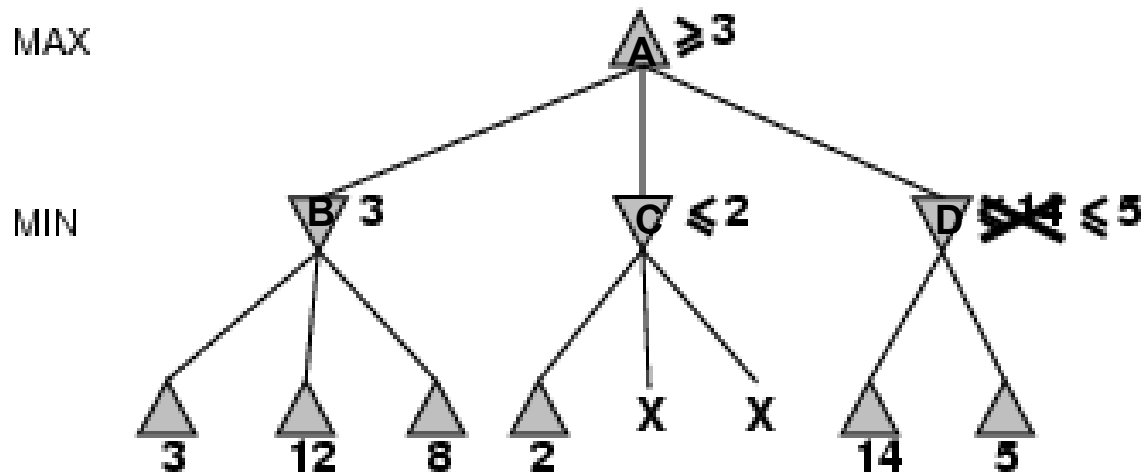


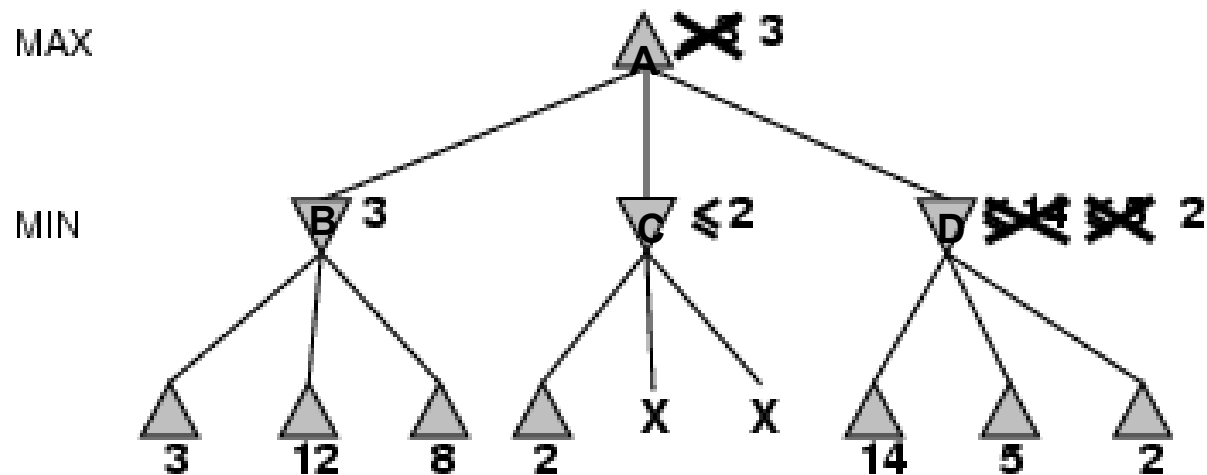
α - β pruning example





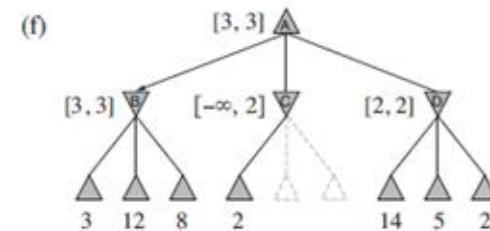
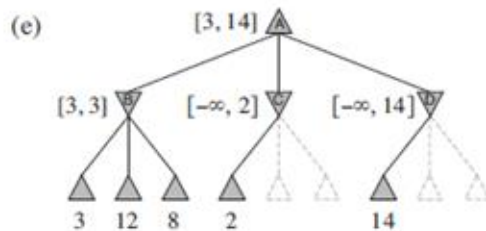
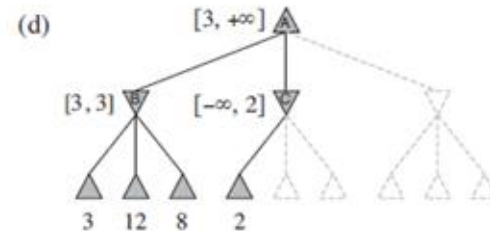
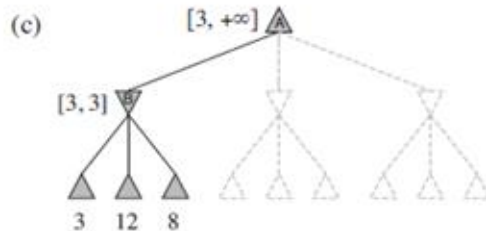
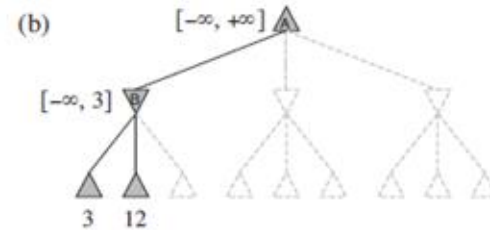
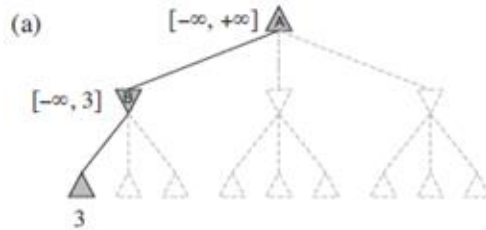




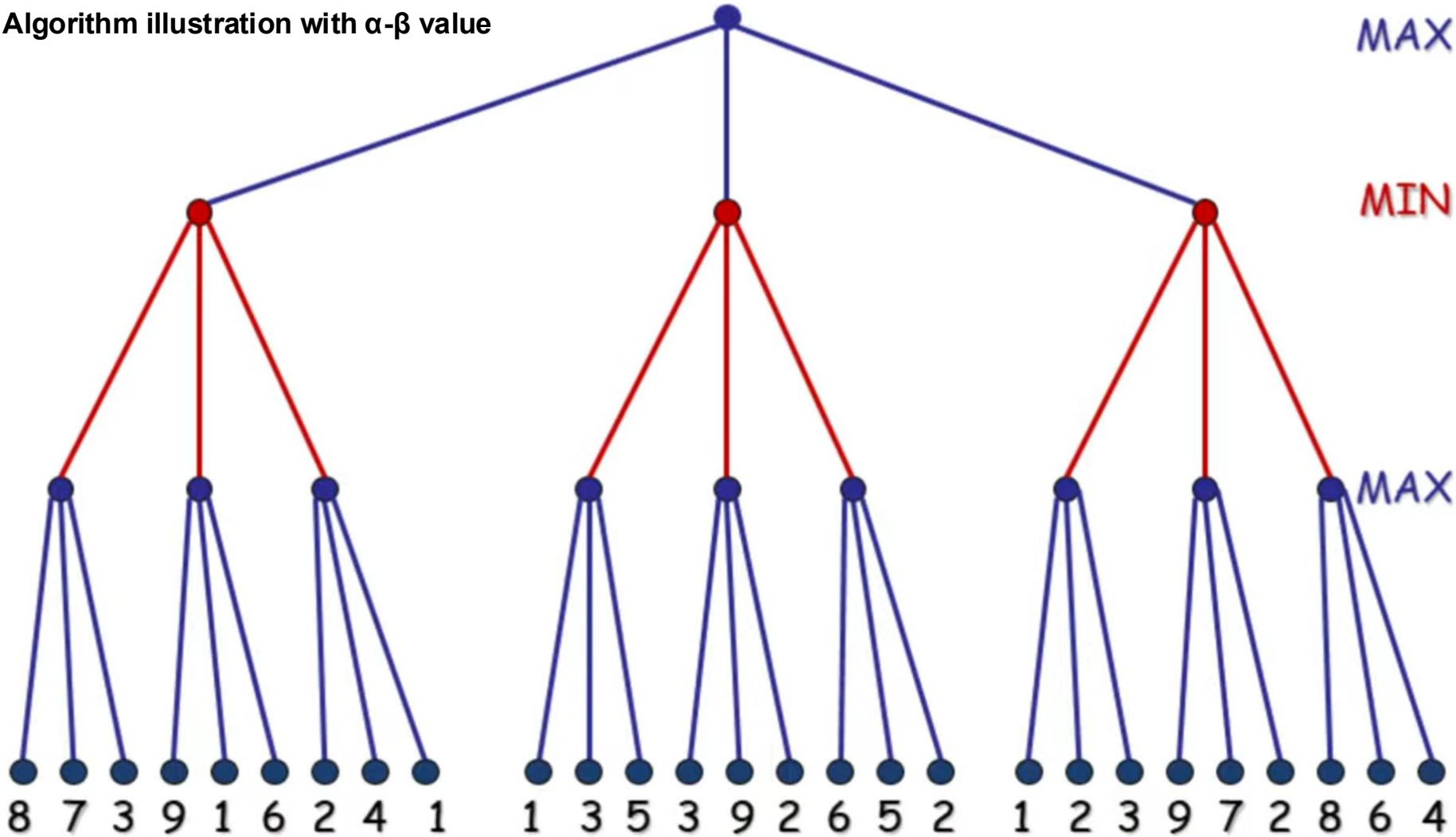


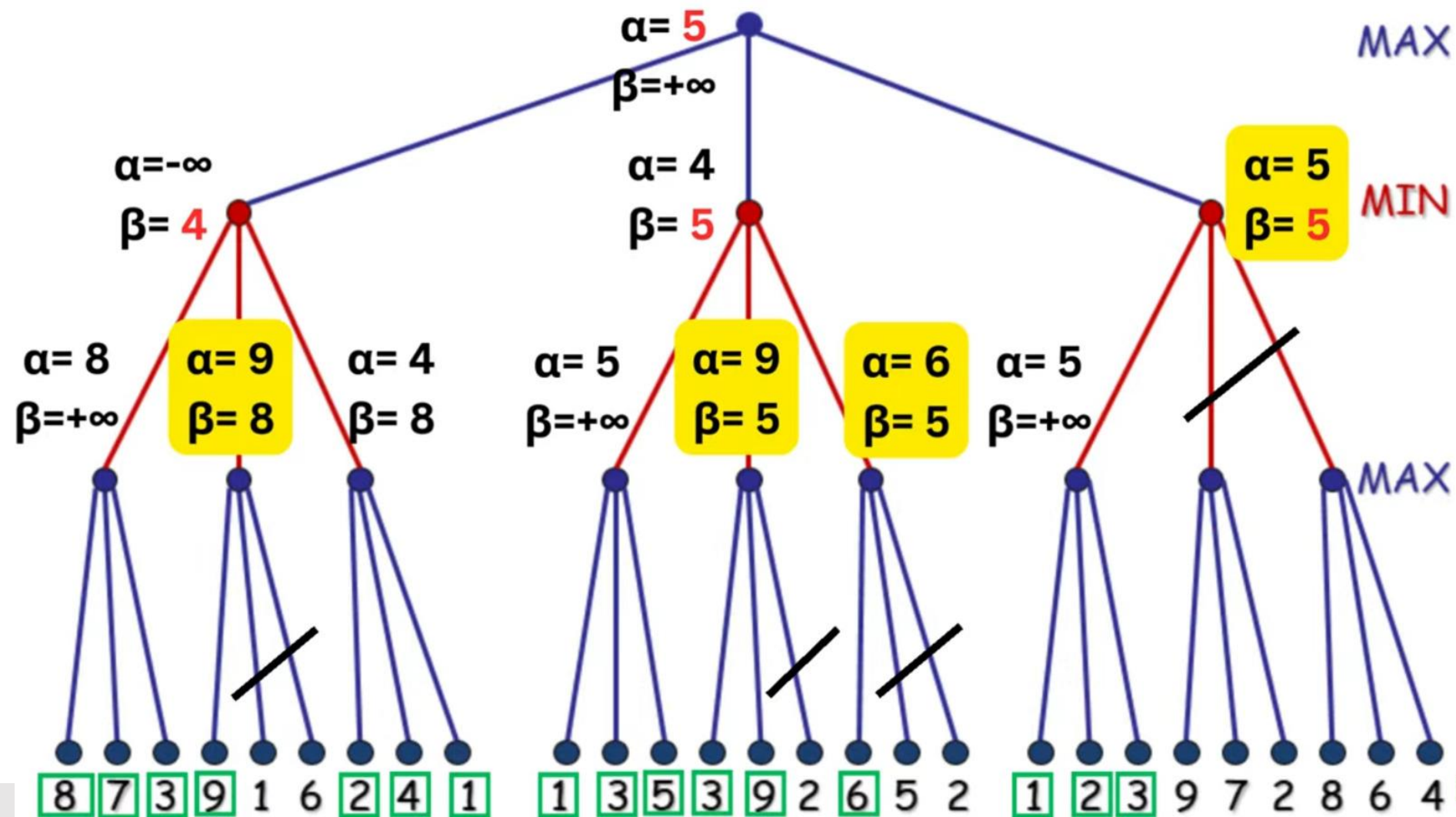


Algorithm illustration with α - β value



Algorithm illustration with α - β value







Adversarial
search

Minimax
search

$\alpha\beta$ Search



Thank You

